Sine, Cosine and Tangent: Ratios Not Sizes

SimuPhysics Right-Triangle Ratio Simulator · SimuPhysics

Year 10	~30 minutes	Mathematics · Geometry & Measurement

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Simulation: https://simuphysics.com/view/mathematics-right-triangle-ratio-simulator.html
Curriculum links: AC9M10M03 · trigonometry

Learning intentions

- I can identify the opposite, adjacent and hypotenuse sides of a right triangle relative to a given angle.
- I can use sine, cosine and tangent to calculate unknown sides and angles in right-angled triangles.
- I can explain why trigonometric ratios depend only on angle, not on the size of the triangle.

Getting started

Open the SimuPhysics Right-Triangle Ratio Simulator. Use the angle slider to change the triangle's angle while the hypotenuse stays fixed. Before revealing each live ratio value, make a prediction — this is where most of the learning happens.

Tasks

1. Set the angle to 30°. Predict whether sin(30°) will be closer to 0 or closer to 1, then reveal the value and check.

2. Increase the angle toward 90°. What happens to the length of the opposite side compared to the hypotenuse? What happens to sin of the angle?

3. Record sin, cos and tan for four different angles of your choice.

Angle	sin(angle)	cos(angle)	tan(angle)

4. Resize the triangle (make it bigger or smaller) while keeping the angle the same. What happens to the side lengths? What happens to the sine, cosine and tangent values? Explain why.

5. A ramp makes a 15° angle with the ground and is 6 m long (the hypotenuse). Use cosine to calculate the horizontal distance it covers.

Working:

6. A ladder leans against a wall at a 70° angle to the ground and reaches 4 m up the wall. Use trigonometry to calculate the length of the ladder.

Working:

7. Explain, using SOH CAH TOA, how you decide which ratio (sine, cosine or tangent) to use when solving a right-triangle problem.

8. Using the simulation, find the angle where sin and cos give exactly the same value. What is special about the triangle at this angle?

Extension (optional)

A surveyor stands 50 m from the base of a cliff and measures the angle of elevation to the top as 38°. Use trigonometry to calculate the height of the cliff, and explain which ratio you chose and why.